

Φάση Ε: Model Evaluation & Σύγκριση

Υπεύθυνος: ML Engineer

Μετρικές:

- Accuracy
- Precision / Recall
- F1-Score
- ROC-AUC
- Confusion Matrix

Οπτικοποιήσεις:

1. Confusion Matrix (heatmap)
2. ROC Curve
3. Συγκριτικά bar charts
4. Πίνακας σύγκρισης

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# Φάση Ε & ΣΤ: Model Evaluation, Σύγκριση & Συμπεράσματα (ΠΛΗΡΗΣ ΚΑΛΥΨΗ + F1-SCORE)
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from pyspark.sql import SparkSession
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import (accuracy_score, precision_score, recall_score,
                             f1_score, confusion_matrix, roc_curve, auc,
                             precision_recall_fscore_support,
                             classification_report)

print("Εκκίνηση SparkSession...")
spark =
SparkSession.builder.appName("Evaluation_Metrics_Final").master("local[*]").getOrCreate()

print("Φόρτωση προβλέψεων (Pandas)...")
df_rf = spark.read.parquet("../data/rf_predictions.parquet").toPandas()
df_svm = spark.read.parquet("../data/svm_predictions.parquet").toPandas()
df_nb = spark.read.parquet("../data/nb_predictions.parquet").toPandas()
df_mlp = spark.read.parquet("../data/mlp_predictions.parquet").toPandas()
df_gbt = spark.read.parquet("../data/gbt_predictions.parquet").toPandas()
df_lr = spark.read.parquet("../data/lr_predictions.parquet").toPandas()

# Λεξικό για πλήρη αποθήκευση μετρικών ανά κλάση και συνολικά
metrics_dict = {
    "Model": [],
    "Acc (Total)": [],
    "P (Class 0)": [], "R (Class 0)": [], "F1 (Class 0)": [],
    "P (Class 1)": [], "R (Class 1)": [], "F1 (Class 1)": [],
    "ROC-AUC": []
}
```

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}

def evaluate_model_full(df, model_name, prob_col="probability"):
    y_true = df['stroke'].values
    y_pred = df['prediction'].values

    # ROC-AUC
    y_prob = np.array([v[1] for v in df[prob_col]]) if prob_col == "probability"
    else np.array([v[1] for v in df[prob_col]])
    fpr, tpr, _ = roc_curve(y_true, y_prob)
    roc_auc = auc(fpr, tpr)

    # Μετρικές ανά κλάση (Precision, Recall, F1)
    precisions, recalls, f1s, _ = precision_recall_fscore_support(y_true, y_pred,
labels=[0, 1], zero_division=0)
    acc = accuracy_score(y_true, y_pred)

    # Προσθήκη στο λεξικό
    metrics_dict["Model"].append(model_name)
    metrics_dict["Acc (Total)"].append(acc)
    metrics_dict["P (Class 0)"].append(precisions[0])
    metrics_dict["R (Class 0)"].append(recalls[0])
    metrics_dict["F1 (Class 0)"].append(f1s[0])
    metrics_dict["P (Class 1)"].append(precisions[1])
    metrics_dict["R (Class 1)"].append(recalls[1])
    metrics_dict["F1 (Class 1)"].append(f1s[1])
    metrics_dict["ROC-AUC"].append(roc_auc)

    cm = confusion_matrix(y_true, y_pred)

    # Εκτύπωση αναλυτικού classification report
    print(f"\n{'='*20} {model_name} {'='*20}")
    print(classification_report(y_true, y_pred, target_names=["Class 0", "Class
1"]))

    return fpr, tpr, roc_auc, cm

# Εκτέλεση αξιολόγησης (Τα 6 Τελικά Μοντέλα μας)
fpr_rf, tpr_rf, auc_rf, cm_rf = evaluate_model_full(df_rf, "Random Forest",
"probability")
fpr_svm, tpr_svm, auc_svm, cm_svm = evaluate_model_full(df_svm, "SVM",
"rawPrediction")
fpr_nb, tpr_nb, auc_nb, cm_nb = evaluate_model_full(df_nb, "Naive Bayes",
"probability")
fpr_mlp, tpr_mlp, auc_mlp, cm_mlp = evaluate_model_full(df_mlp, "Neural
Networks", "probability")
fpr_gbt, tpr_gbt, auc_gbt, cm_gbt = evaluate_model_full(df_gbt, "GBT Classifier",
"probability")
fpr_lr, tpr_lr, auc_lr, cm_lr = evaluate_model_full(df_lr, "Logistic Reg.",

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"probability")

# =====
# 1. ΠΙΝΑΚΑΣ ΣΥΓΚΡΙΣΗΣ (Κεφάλαιο 4: Model Evaluation)
# =====
print("\n" + "="*80)
print("ΠΙΝΑΚΑΣ ΣΥΓΚΡΙΣΗΣ ΜΟΝΤΕΛΩΝ (Απαίτηση Εκφώνησης)")
print("="*80)
comparison_df = pd.DataFrame(metrics_dict).round(3)
print(comparison_df.to_string(index=False))
print("="*80)

# =====
# 2. ΟΠΤΙΚΟΠΟΙΗΣΗ 1: CONFUSION MATRICES (Τέλειο 2x3 Grid)
# =====
fig, axes = plt.subplots(2, 3, figsize=(16, 10))
fig.suptitle('Confusion Matrices (Απαίτηση Εκφώνησης)', fontsize=18,
fontweight='bold')

cms = [cm_rf, cm_svm, cm_nb, cm_mlp, cm_gbt, cm_lr]
titles = ['Random Forest', 'SVM', 'Naive Bayes', 'Neural Networks', 'GBT
Classifier', 'Logistic Reg.']
colors = ['Blues', 'Oranges', 'Greens', 'Purples', 'Reds', 'YlGnBu']

for i in range(len(colors)):
    row, col = i // 3, i % 3
    sns.heatmap(cms[i], annot=True, fmt='d', cmap=colors[i], cbar=False,
ax=axes[row, col])
    axes[row, col].set_title(titles[i], fontsize=12)
    axes[row, col].set_xlabel('Predicted')
    axes[row, col].set_ylabel('True')

plt.tight_layout()
plt.subplots_adjust(top=0.9)
plt.show()

# =====
# 3. ΟΠΤΙΚΟΠΟΙΗΣΗ 2: ROC CURVES
# =====
plt.figure(figsize=(11, 8))
plt.plot(fpr_rf, tpr_rf, label=f'Random Forest (AUC = {auc_rf:.3f})', lw=2)
plt.plot(fpr_svm, tpr_svm, label=f'SVM (AUC = {auc_svm:.3f})', lw=2)
plt.plot(fpr_nb, tpr_nb, label=f'Naive Bayes (AUC = {auc_nb:.3f})', lw=2)
plt.plot(fpr_mlp, tpr_mlp, label=f'Neural Networks (AUC = {auc_mlp:.3f})', lw=2)
plt.plot(fpr_gbt, tpr_gbt, label=f'GBT (AUC = {auc_gbt:.3f})', lw=2,
linestyle='--')
plt.plot(fpr_lr, tpr_lr, label=f'Logistic Reg. (AUC = {auc_lr:.3f})', lw=2,

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linestyle='--')
plt.plot([0, 1], [0, 1], 'k--', lw=2, label='Random Chance')
plt.title('Σύγκριση Μοντέλων: Καμπύλη ROC', fontsize=16, fontweight='bold')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate (Recall)')
plt.legend(loc="lower right")
plt.grid(alpha=0.3)
plt.show()

# =====
# 4. ΟΠΤΙΚΟΠΟΙΗΣΗ 3: ΣΥΓΚΡΙΤΙΚΑ BAR CHARTS
# =====

bar_df = comparison_df.drop(columns=["ROC-AUC"])
metrics_melted = bar_df.melt(id_vars="Model", var_name="Metric",
value_name="Score")

plt.figure(figsize=(14, 7))
sns.barplot(x='Model', y='Score', hue='Metric', data=metrics_melted,
palette='Dark2')
plt.title('Συγκριτικό Διάγραμμα Μετρικών (Accuracy, Precision, Recall, F1-
Score)', fontsize=16, fontweight='bold')
plt.ylim(0, 1.1)
plt.ylabel('Score')
plt.xticks(rotation=30)
plt.legend(title='Μετρικές', bbox_to_anchor=(1.01, 1), loc='upper left')
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.show()

spark.stop()

Εκκίνηση SparkSession...

WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/06/09 23:48:02 WARN Utils: Your hostname, cachyos-x8664, resolves to a
loopback address: 127.0.1.1; using 192.168.1.5 instead (on interface enp4s0)
26/06/09 23:48:02 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another
address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use
setLogLevel(newLevel).
26/06/09 23:48:02 WARN NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable

Φόρτωση προβλέψεων (Pandas)...

===== Random Forest =====
precision    recall  f1-score   support

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Class 0	0.98	0.79	0.88	945
Class 1	0.11	0.57	0.18	42
accuracy			0.78	987
macro avg	0.54	0.68	0.53	987
weighted avg	0.94	0.78	0.85	987

===== SVM =====

	precision	recall	f1-score	support
Class 0	0.98	0.74	0.84	945
Class 1	0.10	0.62	0.17	42
accuracy			0.74	987
macro avg	0.54	0.68	0.51	987
weighted avg	0.94	0.74	0.82	987

===== Naive Bayes =====

	precision	recall	f1-score	support
Class 0	0.97	0.52	0.68	945
Class 1	0.06	0.69	0.11	42
accuracy			0.53	987
macro avg	0.52	0.61	0.39	987
weighted avg	0.94	0.53	0.65	987

===== Neural Networks =====

	precision	recall	f1-score	support
Class 0	0.97	0.81	0.89	945
Class 1	0.09	0.40	0.15	42
accuracy			0.80	987
macro avg	0.53	0.61	0.52	987
weighted avg	0.93	0.80	0.85	987

===== GBT Classifier =====

	precision	recall	f1-score	support
Class 0	0.98	0.79	0.88	945
Class 1	0.12	0.64	0.20	42
accuracy			0.79	987

macro avg	0.55	0.72	0.54	987
weighted avg	0.94	0.79	0.85	987

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===== Logistic Reg. =====
              precision    recall  f1-score   support

   Class 0       0.98       0.74       0.84       945
   Class 1       0.09       0.60       0.16        42

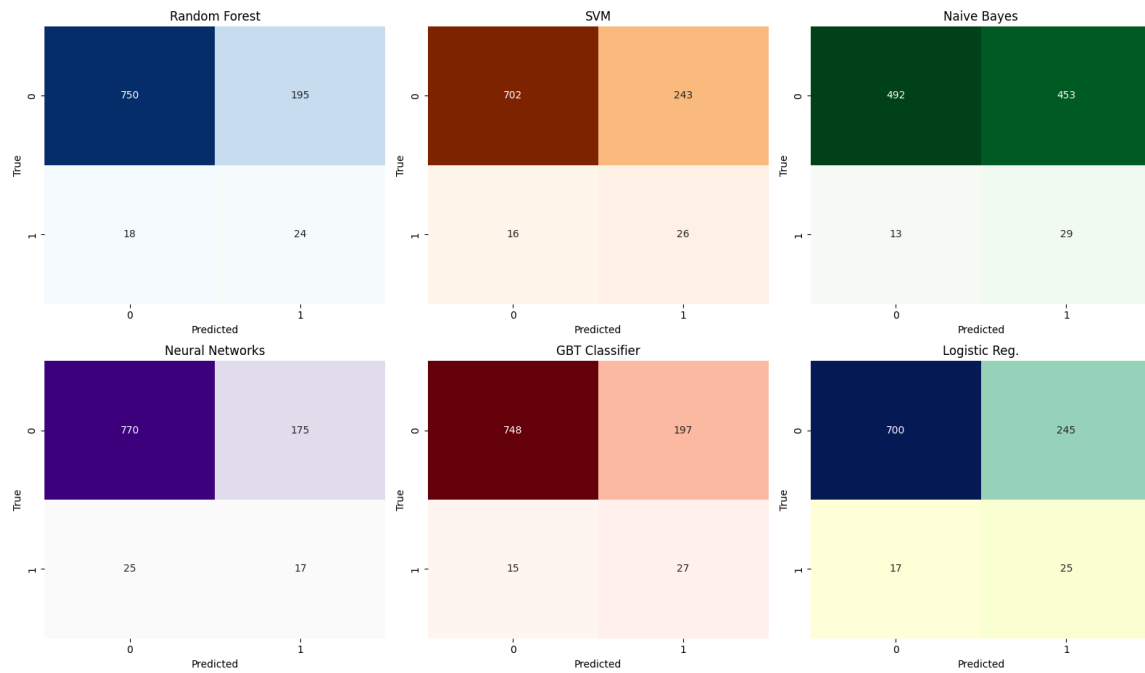
 accuracy              0.73       987
 macro avg              0.53       0.67       0.50       987
 weighted avg           0.94       0.73       0.81       987

```

ΠΙΝΑΚΑΣ ΣΥΓΚΡΙΣΗΣ ΜΟΝΤΕΛΩΝ (Απαίτηση Εκφώνησης)

Model	Acc (Total)	P (Class 0)	R (Class 0)	F1 (Class 0)	P (Class 1)
Random Forest	0.784	0.977	0.794	0.876	0.110
0.571	0.184	0.764			
SVM	0.738	0.978	0.743	0.844	0.097
0.619	0.167	0.761			
Naive Bayes	0.528	0.974	0.521	0.679	0.060
0.690	0.111	0.656			
Neural Networks	0.797	0.969	0.815	0.885	0.089
0.405	0.145	0.727			
GBT Classifier	0.785	0.980	0.792	0.876	0.121
0.643	0.203	0.793			
Logistic Reg.	0.735	0.976	0.741	0.842	0.093
0.595	0.160	0.773			

Confusion Matrices (Απαίτηση Εκφώνησης)



Σύγκριση Μοντέλων: Καμπύλη ROC

